Signal/background separation · mk2 · round 1

```
date         2026-10-25 16:44
data         C:\Users\natha\OneDrive\Desktop\0_mL_ATLAS\0_Sig_BG_09_21_KJ\kj_092126\run3_csv_for_nathan
background   DataHTMissLow_train_data24_cleaned.csv  rows 36,059
signal       HSS_mH125_mS15_ct580_cleaned.csv  rows 570
signal       HSS_mH125_mS55_ct5320_cleaned.csv  rows 423
signal       HSS_mH200_mS50_ct1255_cleaned.csv  rows 1,520
signal       HSS_mH400_mS100_ct1610_cleaned.csv  rows 3,792
signal       HSS_mH600_mS150_ct3310_cleaned.csv  rows 4,490
signal       HSS_mH1000_mS275_ct2399_cleaned.csv  rows 4,188

selections   all  = every event
             ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
region       ('msvtx_inBarrel_NOSYS', 'flag', 'msvtx_inEndcaps_NOSYS')

features common to all 7 files   230
hard / user excluded             15 / 0
non-numeric / constant overall   6 / 20
exact duplicates dropped         20  (in 20 groups; duplicates.txt)
features scored                  169
constant in >= 1 cell            26  (no page in those cells)

metric   two-sample KS D, unweighted events, NaN and -999 removed per feature.
         D_equal weights each mass point 1/N_m; D_pooled concatenates them.
folders  features/NNN_<feature>/, NNN = rank by endcap / ht40 / D_equal

this PDF: cover, selection report, then per selection and region a ranking chart and
top-40 table, then per region the 40 largest drops from all to ht40.
```

Selection report (1/2)

```
cut variable   htmissScalar_NOSYS
threshold      40 GeV  =  40000 in file units
units          MeV  (signal median 229,019.4 file units; rule: median > 1000 -> MeV)

per-file edges, in GeV assuming the same units for every listed variable:
  file          variable                  min     q0.1%       q1%    median   % < cut
  bkg           htmissScalar_NOSYS        0.00      0.00      0.00     26.69    78.90%
  mH125_mS15    htmissScalar_NOSYS       40.76     41.09     43.13    126.41     0.00%
  mH125_mS55    htmissScalar_NOSYS       40.05     40.20     42.59    162.50     0.00%
  mH200_mS50    htmissScalar_NOSYS       40.08     40.14     42.92    127.62     0.00%
  mH400_mS100   htmissScalar_NOSYS       40.20     41.49     48.66    183.64     0.00%
  mH600_mS150   htmissScalar_NOSYS       40.03     40.43     44.31    226.81     0.00%
  mH1000_mS275  htmissScalar_NOSYS       40.49     40.81     47.26    388.57     0.00%

  bkg           htmiss_NOSYS              0.00      0.00      0.00     18.94
  mH125_mS15    htmiss_NOSYS             40.36     40.37     41.47     86.20
  mH125_mS55    htmiss_NOSYS             40.05     40.10     40.74    125.69
  mH200_mS50    htmiss_NOSYS             40.08     40.14     41.05     91.18
  mH400_mS100   htmiss_NOSYS             40.20     40.58     44.34    132.91
  mH600_mS150   htmiss_NOSYS             40.00     40.22     41.96    164.49
  mH1000_mS275  htmiss_NOSYS             40.02     40.77     44.75    279.53

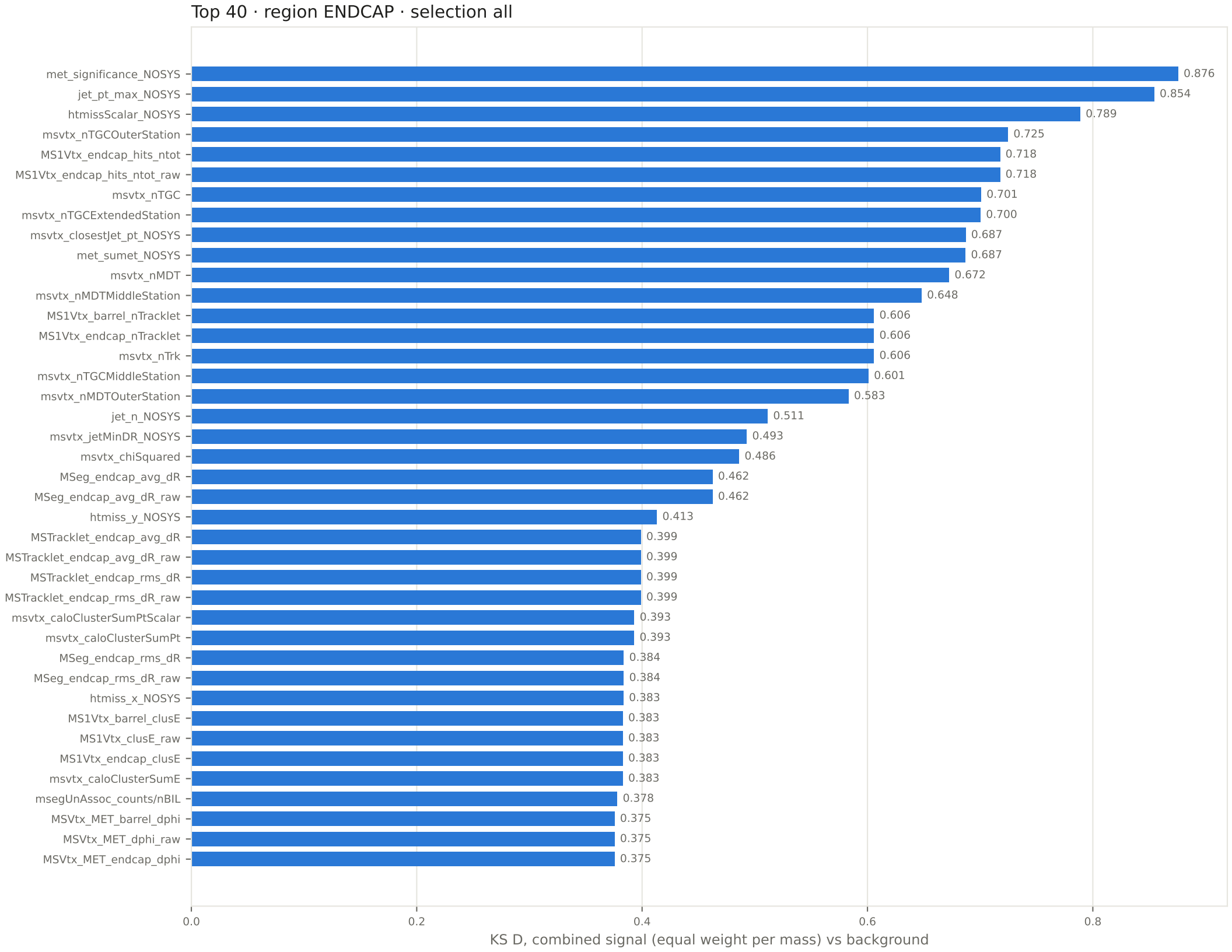
  bkg           met_met_NOSYS             0.00      0.42      1.53     18.91
  mH125_mS15    met_met_NOSYS            21.55     22.50     24.28     88.31
  mH125_mS55    met_met_NOSYS            23.10     23.28     35.81    130.93
  mH200_mS50    met_met_NOSYS            20.08     22.03     29.15     92.50
  mH400_mS100   met_met_NOSYS            20.07     22.43     35.05    132.38
  mH600_mS150   met_met_NOSYS            20.26     21.59     32.90    164.70
  mH1000_mS275  met_met_NOSYS            22.32     25.21     39.91    279.95

  bkg           jet_pt_max_NOSYS          0.00      0.00      0.00     26.58
  mH125_mS15    jet_pt_max_NOSYS          0.00      0.00     26.46     75.00
  mH125_mS55    jet_pt_max_NOSYS         28.41     29.45     34.30    106.77
  mH200_mS50    jet_pt_max_NOSYS         25.75     27.46     31.08     81.56
  mH400_mS100   jet_pt_max_NOSYS         27.04     28.64     35.26    117.68
  mH600_mS150   jet_pt_max_NOSYS         25.85     28.46     34.42    140.79
  mH1000_mS275  jet_pt_max_NOSYS         26.68     30.42     38.95    231.53

events per file and region, before and after the cut:
  file          region       all     ht40    kept    below   no HT
  bkg           endcap    34,970    7,390   21.1%   27,580       0
  bkg           barrel     1,089      220   20.2%      869       0
  bkg           all       36,059    7,610   21.1%   28,449       0
  mH125_mS15    endcap       411      411  100.0%        0       0
  mH125_mS15    barrel       159      159  100.0%        0       0
  mH125_mS15    all          570      570  100.0%        0       0
  mH125_mS55    endcap       281      281  100.0%        0       0
  mH125_mS55    barrel       142      142  100.0%        0       0
  mH125_mS55    all          423      423  100.0%        0       0
  mH200_mS50    endcap     1,071    1,071  100.0%        0       0
  mH200_mS50    barrel       449      449  100.0%        0       0
  mH200_mS50    all        1,520    1,520  100.0%        0       0
  mH400_mS100   endcap     2,648    2,648  100.0%        0       0
  mH400_mS100   barrel     1,144    1,144  100.0%        0       0
  mH400_mS100   all        3,792    3,792  100.0%        0       0
  mH600_mS150   endcap     2,882    2,882  100.0%        0       0
  mH600_mS150   barrel     1,608    1,608  100.0%        0       0
  mH600_mS150   all        4,490    4,490  100.0%        0       0
  mH1000_mS275  endcap     2,703    2,703  100.0%        0       0
  mH1000_mS275  barrel     1,485    1,485  100.0%        0       0
  mH1000_mS275  all        4,188    4,188  100.0%        0       0
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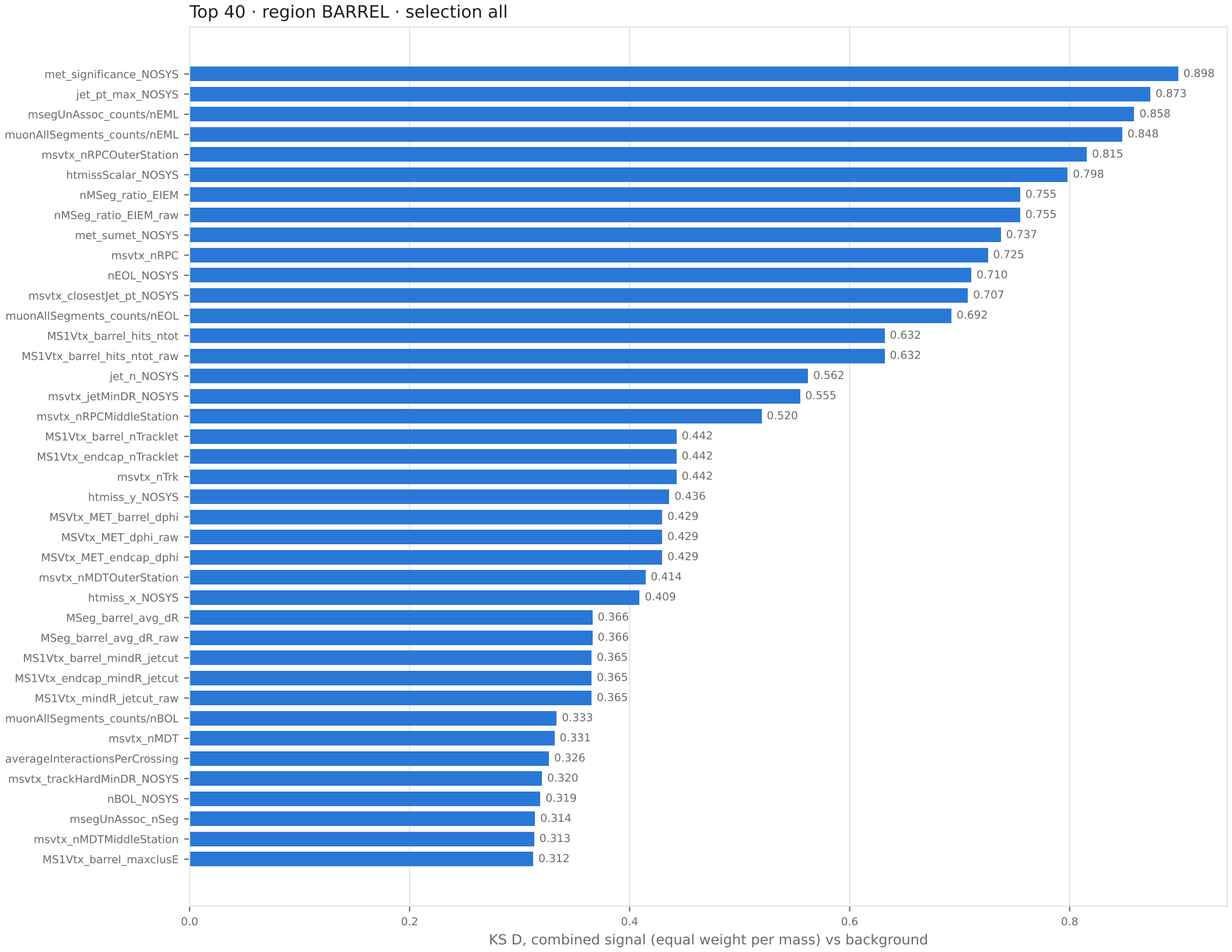
Selection report (2/2)

all signal files keep >= 99% of events after the cut


Top 40 · region ENDCAP · selection all
met_significance_NOSYS 0.876
jet_pt_max_NOSYS 0.854
htmissScalar_NOSYS 0.789
msvtx_nTGCOuterStation 0.725
MS1Vtx_endcap_hits_ntot 0.718
MS1Vtx_endcap_hits_ntot_raw 0.718
msvtx_nTGC 0.701
msvtx_nTGCExtendedStation 0.700
msvtx_closestJet_pt_NOSYS 0.687
met_sumet_NOSYS 0.687
msvtx_nMDT 0.672
msvtx_nMDTMiddleStation 0.648
MS1Vtx_barrel_nTracklet 0.606
MS1Vtx_endcap_nTracklet 0.606
msvtx_nTrk 0.606
msvtx_nTGCMiddleStation 0.601
msvtx_nMDTOuterStation 0.583
jet_n_NOSYS 0.511
msvtx_jetMinDR_NOSYS 0.493
msvtx_chiSquared 0.486
MSeg_endcap_avg_dR 0.462
MSeg_endcap_avg_dR_raw 0.462
htmiss_y_NOSYS 0.413
MSTracklet_endcap_avg_dR 0.399
MSTracklet_endcap_avg_dR_raw 0.399
MSTracklet_endcap_rms_dR 0.399
MSTracklet_endcap_rms_dR_raw 0.399
msvtx_caloClusterSumPtScalar 0.393
msvtx_caloClusterSumPt 0.393
MSeg_endcap_rms_dR 0.384
MSeg_endcap_rms_dR_raw 0.384
htmiss_x_NOSYS 0.383
MS1Vtx_barrel_clusE 0.383
MS1Vtx_clusE_raw 0.383
MS1Vtx_endcap_clusE 0.383
msvtx_caloClusterSumE 0.383
msegUnAssoc_counts/nBIL 0.378
MSVtx_MET_barrel_dphi 0.375
MSVtx_MET_dphi_raw 0.375
MSVtx_MET_endcap_dphi 0.375
0.0
0.2
0.4
0.6
0.8
KS D, combined signal (equal weight per mass) vs background

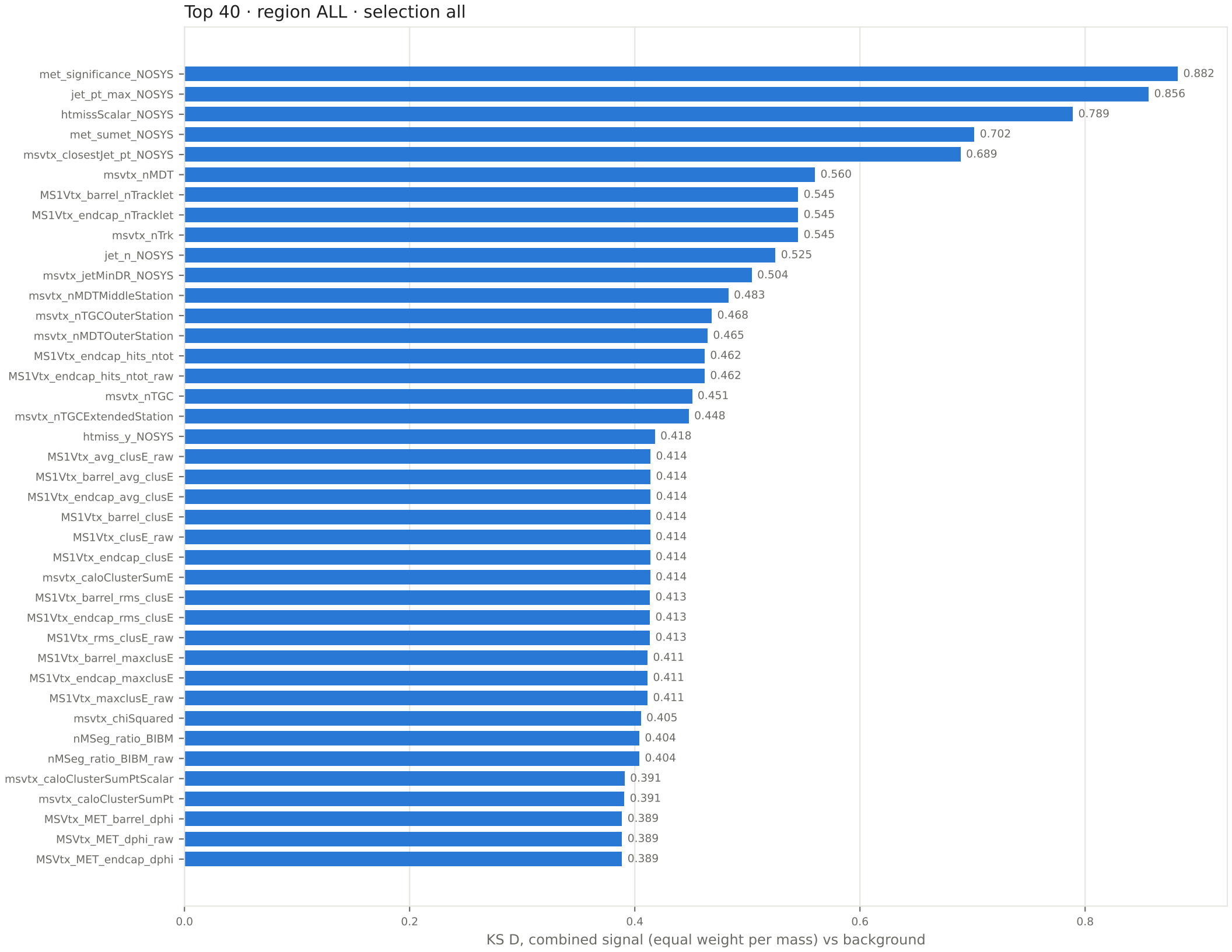
Top 40 table · region ENDCAP · selection all

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	met_significance_NOSYS	0.876	0.896	0.810	0.939	34,970
2	jet_pt_max_NOSYS	0.854	0.874	0.785	0.920	34,970
3	htmissScalar_NOSYS	0.789	0.789	0.789	0.868	34,970
4	msvtx_nTGCOuterStation	0.725	0.785	0.612	0.839	34,970
5	MS1Vtx_endcap_hits_ntot	0.718	0.794	0.553	0.864	34,970
6	MS1Vtx_endcap_hits_ntot_raw	0.718	0.794	0.553	0.864	34,970
7	msvtx_nTGC	0.701	0.773	0.590	0.840	34,970
8	msvtx_nTGCExtendedStation	0.700	0.766	0.585	0.828	34,970
9	msvtx_closestJet_pt_NOSYS	0.687	0.697	0.647	0.715	34,970
10	met_sumet_NOSYS	0.687	0.741	0.563	0.859	34,970
11	msvtx_nMDT	0.672	0.759	0.491	0.846	34,970
12	msvtx_nMDTMiddleStation	0.648	0.736	0.504	0.827	34,970
13	MS1Vtx_barrel_nTracklet	0.606	0.607	0.551	0.643	34,970
14	MS1Vtx_endcap_nTracklet	0.606	0.607	0.551	0.643	34,970
15	msvtx_nTrk	0.606	0.607	0.551	0.643	34,970
16	msvtx_nTGCMiddleStation	0.601	0.683	0.471	0.755	34,970
17	msvtx_nMDTOuterStation	0.583	0.636	0.465	0.686	34,970
18	jet_n_NOSYS	0.511	0.556	0.423	0.647	34,970
19	msvtx_jetMinDR_NOSYS	0.493	0.512	0.468	0.538	34,970
20	msvtx_chiSquared	0.486	0.576	0.582	1.000	34,970
21	MSeg_endcap_avg_dR	0.462	0.541	0.235	0.667	34,970
22	MSeg_endcap_avg_dR_raw	0.462	0.541	0.235	0.667	34,970
23	htmiss_y_NOSYS	0.413	0.417	0.368	0.449	34,970
24	MSTracklet_endcap_avg_dR	0.399	0.431	0.297	0.502	34,970
25	MSTracklet_endcap_avg_dR_raw	0.399	0.431	0.297	0.502	34,970
26	MSTracklet_endcap_rms_dR	0.399	0.431	0.297	0.502	34,970
27	MSTracklet_endcap_rms_dR_raw	0.399	0.431	0.297	0.502	34,970
28	msvtx_caloClusterSumPtScalar	0.393	0.394	0.389	0.404	34,970
29	msvtx_caloClusterSumPt	0.393	0.393	0.388	0.405	34,970
30	MSeg_endcap_rms_dR	0.384	0.457	0.180	0.594	34,970
31	MSeg_endcap_rms_dR_raw	0.384	0.457	0.180	0.594	34,970
32	htmiss_x_NOSYS	0.383	0.398	0.354	0.438	34,970
33	MS1Vtx_barrel_clusE	0.383	0.386	0.372	0.396	34,970
34	MS1Vtx_clusE_raw	0.383	0.386	0.372	0.396	34,970
35	MS1Vtx_endcap_clusE	0.383	0.386	0.372	0.396	34,970
36	msvtx_caloClusterSumE	0.383	0.386	0.372	0.396	34,970
37	msegUnAssoc_counts/nBIL	0.378	0.346	0.314	0.447	34,970
38	MSVtx_MET_barrel_dphi	0.375	0.357	0.273	0.508	34,970
39	MSVtx_MET_dphi_raw	0.375	0.357	0.273	0.508	34,970
40	MSVtx_MET_endcap_dphi	0.375	0.357	0.273	0.508	34,970


Top 40 · region BARREL · selection all
met_significance_NOSYS 0.898
jet_pt_max_NOSYS 0.873
msegUnAssoc_counts/nEML 0.858
muonAllSegments_counts/nEML 0.848
msvtx_nRPCOuterStation 0.815
htmissScalar_NOSYS 0.798
nMSeg_ratio_EIEM 0.755
nMSeg_ratio_EIEM_raw 0.755
met_sumet_NOSYS 0.737
msvtx_nRPC 0.725
nEOL_NOSYS 0.710
msvtx_closestJet_pt_NOSYS 0.707
muonAllSegments_counts/nEOL 0.692
MS1Vtx_barrel_hits_ntot 0.632
MS1Vtx_barrel_hits_ntot_raw 0.632
jet_n_NOSYS 0.562
msvtx_jetMinDR_NOSYS 0.555
msvtx_nRPCMiddleStation 0.520
MS1Vtx_barrel_nTracklet 0.442
MS1Vtx_endcap_nTracklet 0.442
msvtx_nTrk 0.442
htmiss_y_NOSYS 0.436
MSVtx_MET_barrel_dphi 0.429
MSVtx_MET_dphi_raw 0.429
MSVtx_MET_endcap_dphi 0.429
msvtx_nMDTOuterStation 0.414
htmiss_x_NOSYS 0.409
MSeg_barrel_avg_dR 0.366
MSeg_barrel_avg_dR_raw 0.366
MS1Vtx_barrel_mindR_jetcut 0.365
MS1Vtx_endcap_mindR_jetcut 0.365
MS1Vtx_mindR_jetcut_raw 0.365
muonAllSegments_counts/nBOL 0.333
msvtx_nMDT 0.331
averageInteractionsPerCrossing 0.326
msvtx_trackHardMinDR_NOSYS 0.320
nBOL_NOSYS 0.319
msegUnAssoc_nSeg 0.314
msvtx_nMDTMiddleStation 0.313
MS1Vtx_barrel_maxclusE 0.312
0.0
0.2
0.4
0.6
0.8
KS D, combined signal (equal weight per mass) vs background

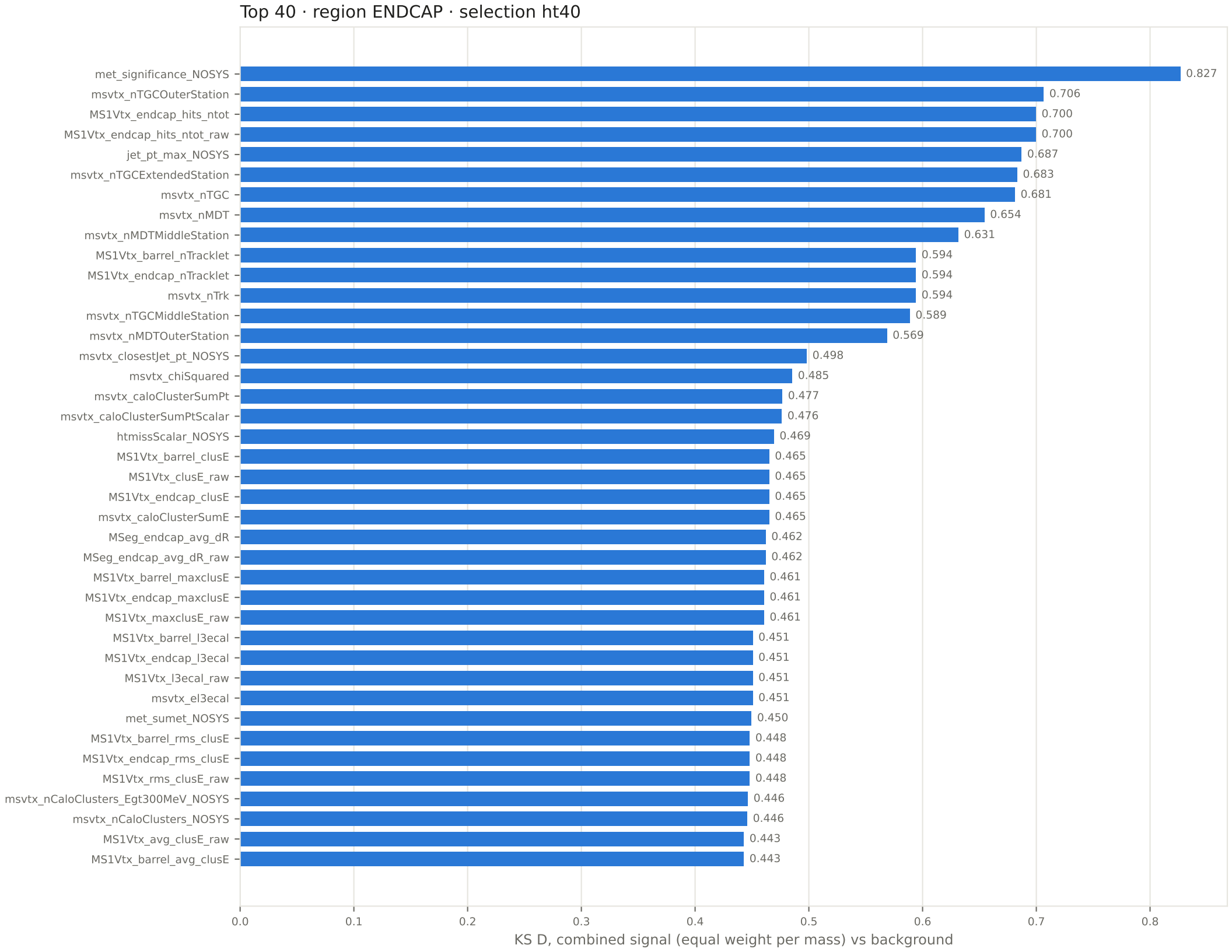
Top 40 table · region BARREL · selection all

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	met_significance_NOSYS	0.898	0.919	0.831	0.962	1,089
2	jet_pt_max_NOSYS	0.873	0.892	0.789	0.916	1,089
3	msegUnAssoc_counts/nEML	0.858	0.843	0.821	0.894	1,089
4	muonAllSegments_counts/nEML	0.848	0.828	0.799	0.901	1,089
5	msvtx_nRPCOuterStation	0.815	0.843	0.740	0.881	1,089
6	htmissScalar_NOSYS	0.798	0.818	0.798	0.872	1,089
7	nMSeg_ratio_EIEM	0.755	0.746	0.727	0.805	1,089
8	nMSeg_ratio_EIEM_raw	0.755	0.746	0.727	0.805	1,089
9	met_sumet_NOSYS	0.737	0.783	0.588	0.852	1,089
10	msvtx_nRPC	0.725	0.787	0.647	0.850	1,089
11	nEOL_NOSYS	0.710	0.651	0.580	0.844	1,089
12	msvtx_closestJet_pt_NOSYS	0.707	0.728	0.639	0.740	1,089
13	muonAllSegments_counts/nEOL	0.692	0.632	0.563	0.821	1,089
14	MS1Vtx_barrel_hits_ntot	0.632	0.727	0.405	0.834	1,089
15	MS1Vtx_barrel_hits_ntot_raw	0.632	0.727	0.405	0.834	1,089
16	jet_n_NOSYS	0.562	0.611	0.458	0.691	1,089
17	msvtx_jetMinDR_NOSYS	0.555	0.583	0.510	0.622	1,089
18	msvtx_nRPCMiddleStation	0.520	0.611	0.314	0.691	1,089
19	MS1Vtx_barrel_nTracklet	0.442	0.458	0.418	0.475	1,089
20	MS1Vtx_endcap_nTracklet	0.442	0.458	0.418	0.475	1,089
21	msvtx_nTrk	0.442	0.458	0.418	0.475	1,089
22	htmiss_y_NOSYS	0.436	0.437	0.374	0.470	1,089
23	MSVtx_MET_barrel_dphi	0.429	0.407	0.332	0.620	1,089
24	MSVtx_MET_dphi_raw	0.429	0.407	0.332	0.620	1,089
25	MSVtx_MET_endcap_dphi	0.429	0.407	0.332	0.620	1,089
26	msvtx_nMDTOuterStation	0.414	0.565	0.080	0.710	1,089
27	htmiss_x_NOSYS	0.409	0.421	0.363	0.458	1,089
28	MSeg_barrel_avg_dR	0.366	0.497	0.101	0.645	1,089
29	MSeg_barrel_avg_dR_raw	0.366	0.497	0.101	0.645	1,089
30	MS1Vtx_barrel_mindR_jetcut	0.365	0.391	0.307	0.432	1,089
31	MS1Vtx_endcap_mindR_jetcut	0.365	0.391	0.307	0.432	1,089
32	MS1Vtx_mindR_jetcut_raw	0.365	0.391	0.307	0.432	1,089
33	muonAllSegments_counts/nBOL	0.333	0.425	0.151	0.517	1,089
34	msvtx_nMDT	0.331	0.500	0.161	0.687	1,089
35	averageInteractionsPerCrossing	0.326	0.324	0.307	0.367	1,089
36	msvtx_trackHardMinDR_NOSYS	0.320	0.346	0.273	0.393	1,089
37	nBOL_NOSYS	0.319	0.408	0.142	0.508	1,089
38	msegUnAssoc_nSeg	0.314	0.175	0.113	0.652	1,089
39	msvtx_nMDTMiddleStation	0.313	0.444	0.196	0.573	1,089
40	MS1Vtx_barrel_maxclusE	0.312	0.314	0.275	0.332	1,089


Top 40 · region ALL · selection all
met_significance_NOSYS 0.882
jet_pt_max_NOSYS 0.856
htmissScalar_NOSYS 0.789
met_sumet_NOSYS 0.702
msvtx_closestJet_pt_NOSYS 0.689
msvtx_nMDT 0.560
MS1Vtx_barrel_nTracklet 0.545
MS1Vtx_endcap_nTracklet 0.545
msvtx_nTrk 0.545
jet_n_NOSYS 0.525
msvtx_jetMinDR_NOSYS 0.504
msvtx_nMDTMiddleStation 0.483
msvtx_nTGCOuterStation 0.468
msvtx_nMDTOuterStation 0.465
MS1Vtx_endcap_hits_ntot 0.462
MS1Vtx_endcap_hits_ntot_raw 0.462
msvtx_nTGC 0.451
msvtx_nTGCExtendedStation 0.448
htmiss_y_NOSYS 0.418
MS1Vtx_avg_clusE_raw 0.414
MS1Vtx_barrel_avg_clusE 0.414
MS1Vtx_endcap_avg_clusE 0.414
MS1Vtx_barrel_clusE 0.414
MS1Vtx_clusE_raw 0.414
MS1Vtx_endcap_clusE 0.414
msvtx_caloClusterSumE 0.414
MS1Vtx_barrel_rms_clusE 0.413
MS1Vtx_endcap_rms_clusE 0.413
MS1Vtx_rms_clusE_raw 0.413
MS1Vtx_barrel_maxclusE 0.411
MS1Vtx_endcap_maxclusE 0.411
MS1Vtx_maxclusE_raw 0.411
msvtx_chiSquared 0.405
nMSeg_ratio_BIBM 0.404
nMSeg_ratio_BIBM_raw 0.404
msvtx_caloClusterSumPtScalar 0.391
msvtx_caloClusterSumPt 0.391
MSVtx_MET_barrel_dphi 0.389
MSVtx_MET_dphi_raw 0.389
MSVtx_MET_endcap_dphi 0.389
0.0
0.2
0.4
0.6
0.8
KS D, combined signal (equal weight per mass) vs background

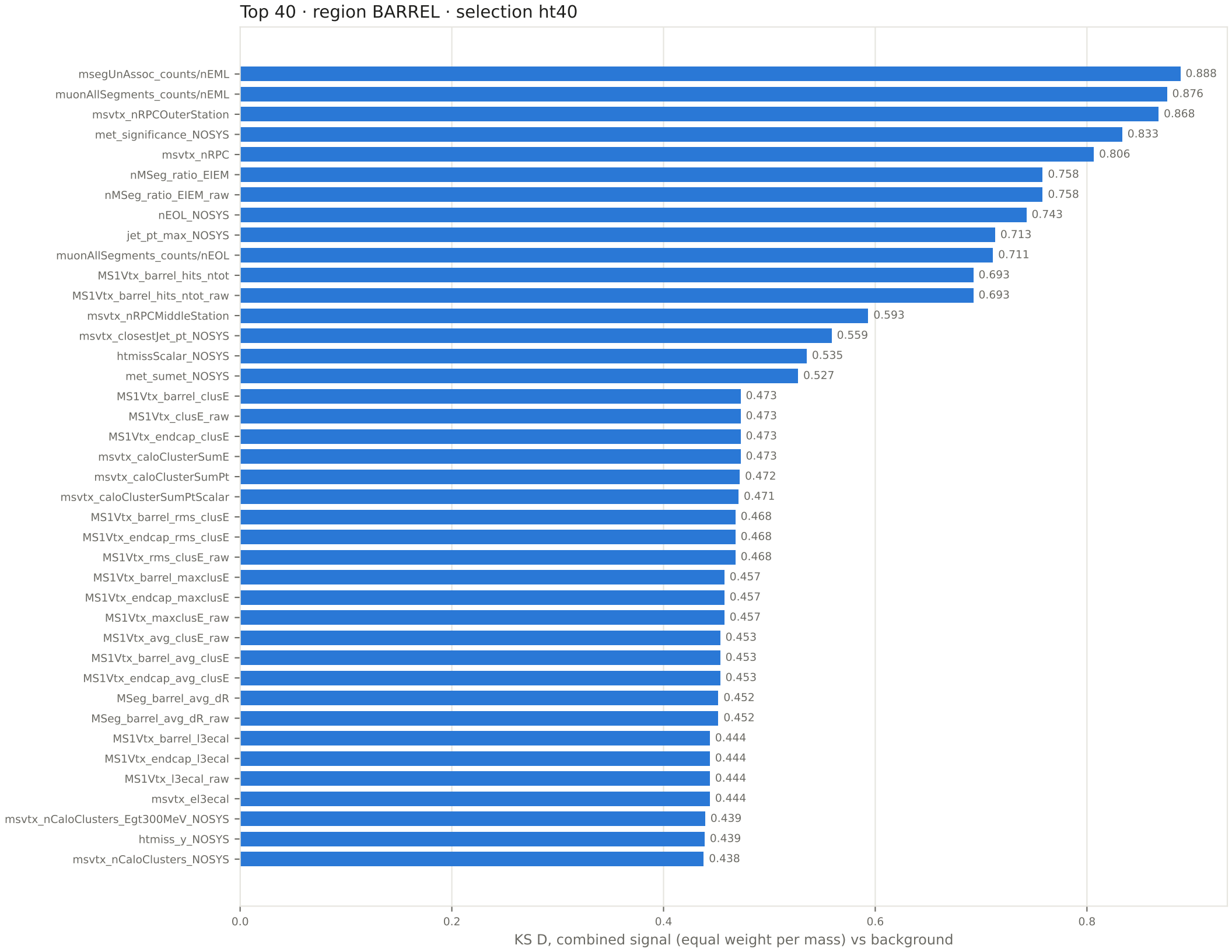
Top 40 table · region ALL · selection all

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	met_significance_NOSYS	0.882	0.902	0.814	0.938	36,059
2	jet_pt_max_NOSYS	0.856	0.876	0.782	0.913	36,059
3	htmissScalar_NOSYS	0.789	0.792	0.789	0.868	36,059
4	met_sumet_NOSYS	0.702	0.753	0.563	0.856	36,059
5	msvtx_closestJet_pt_NOSYS	0.689	0.703	0.640	0.719	36,059
6	msvtx_nMDT	0.560	0.666	0.366	0.783	36,059
7	MS1Vtx_barrel_nTracklet	0.545	0.548	0.502	0.569	36,059
8	MS1Vtx_endcap_nTracklet	0.545	0.548	0.502	0.569	36,059
9	msvtx_nTrk	0.545	0.548	0.502	0.569	36,059
10	jet_n_NOSYS	0.525	0.572	0.422	0.659	36,059
11	msvtx_jetMinDR_NOSYS	0.504	0.525	0.480	0.548	36,059
12	msvtx_nMDTMiddleStation	0.483	0.573	0.321	0.670	36,059
13	msvtx_nTGCOuterStation	0.468	0.505	0.371	0.532	36,059
14	msvtx_nMDTOuterStation	0.465	0.543	0.326	0.634	36,059
15	MS1Vtx_endcap_hits_ntot	0.462	0.511	0.335	0.547	36,059
16	MS1Vtx_endcap_hits_ntot_raw	0.462	0.511	0.335	0.547	36,059
17	msvtx_nTGC	0.451	0.498	0.354	0.527	36,059
18	msvtx_nTGCExtendedStation	0.448	0.490	0.351	0.519	36,059
19	htmiss_y_NOSYS	0.418	0.422	0.364	0.447	36,059
20	MS1Vtx_avg_clusE_raw	0.414	0.421	0.396	0.429	36,059
21	MS1Vtx_barrel_avg_clusE	0.414	0.421	0.396	0.429	36,059
22	MS1Vtx_endcap_avg_clusE	0.414	0.421	0.396	0.429	36,059
23	MS1Vtx_barrel_clusE	0.414	0.415	0.411	0.420	36,059
24	MS1Vtx_clusE_raw	0.414	0.415	0.411	0.420	36,059
25	MS1Vtx_endcap_clusE	0.414	0.415	0.411	0.420	36,059
26	msvtx_caloClusterSumE	0.414	0.415	0.411	0.420	36,059
27	MS1Vtx_barrel_rms_clusE	0.413	0.420	0.399	0.428	36,059
28	MS1Vtx_endcap_rms_clusE	0.413	0.420	0.399	0.428	36,059
29	MS1Vtx_rms_clusE_raw	0.413	0.420	0.399	0.428	36,059
30	MS1Vtx_barrel_maxclusE	0.411	0.414	0.394	0.419	36,059
31	MS1Vtx_endcap_maxclusE	0.411	0.414	0.394	0.419	36,059
32	MS1Vtx_maxclusE_raw	0.411	0.414	0.394	0.419	36,059
33	msvtx_chiSquared	0.405	0.490	0.494	1.000	36,059
34	nMSeg_ratio_BIBM	0.404	0.410	0.381	0.427	36,059
35	nMSeg_ratio_BIBM_raw	0.404	0.410	0.381	0.427	36,059
36	msvtx_caloClusterSumPtScalar	0.391	0.392	0.378	0.403	36,059
37	msvtx_caloClusterSumPt	0.391	0.392	0.377	0.403	36,059
38	MSVtx_MET_barrel_dphi	0.389	0.368	0.292	0.536	36,059
39	MSVtx_MET_dphi_raw	0.389	0.368	0.292	0.536	36,059
40	MSVtx_MET_endcap_dphi	0.389	0.368	0.292	0.536	36,059


Top 40 · region ENDCAP · selection ht40
met_significance_NOSYS 0.827
msvtx_nTGCOuterStation 0.706
MS1Vtx_endcap_hits_ntot 0.700
MS1Vtx_endcap_hits_ntot_raw 0.700
jet_pt_max_NOSYS 0.687
msvtx_nTGCExtendedStation 0.683
msvtx_nTGC 0.681
msvtx_nMDT 0.654
msvtx_nMDTMiddleStation 0.631
MS1Vtx_barrel_nTracklet 0.594
MS1Vtx_endcap_nTracklet 0.594
msvtx_nTrk 0.594
msvtx_nTGCMiddleStation 0.589
msvtx_nMDTOuterStation 0.569
msvtx_closestJet_pt_NOSYS 0.498
msvtx_chiSquared 0.485
msvtx_caloClusterSumPt 0.477
msvtx_caloClusterSumPtScalar 0.476
htmissScalar_NOSYS 0.469
MS1Vtx_barrel_clusE 0.465
MS1Vtx_clusE_raw 0.465
MS1Vtx_endcap_clusE 0.465
msvtx_caloClusterSumE 0.465
MSeg_endcap_avg_dR 0.462
MSeg_endcap_avg_dR_raw 0.462
MS1Vtx_barrel_maxclusE 0.461
MS1Vtx_endcap_maxclusE 0.461
MS1Vtx_maxclusE_raw 0.461
MS1Vtx_barrel_l3ecal 0.451
MS1Vtx_endcap_l3ecal 0.451
MS1Vtx_l3ecal_raw 0.451
msvtx_el3ecal 0.451
met_sumet_NOSYS 0.450
MS1Vtx_barrel_rms_clusE 0.448
MS1Vtx_endcap_rms_clusE 0.448
MS1Vtx_rms_clusE_raw 0.448
msvtx_nCaloClusters_Egt300MeV_NOSYS 0.446
msvtx_nCaloClusters_NOSYS 0.446
MS1Vtx_avg_clusE_raw 0.443
MS1Vtx_barrel_avg_clusE 0.443
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
KS D, combined signal (equal weight per mass) vs background

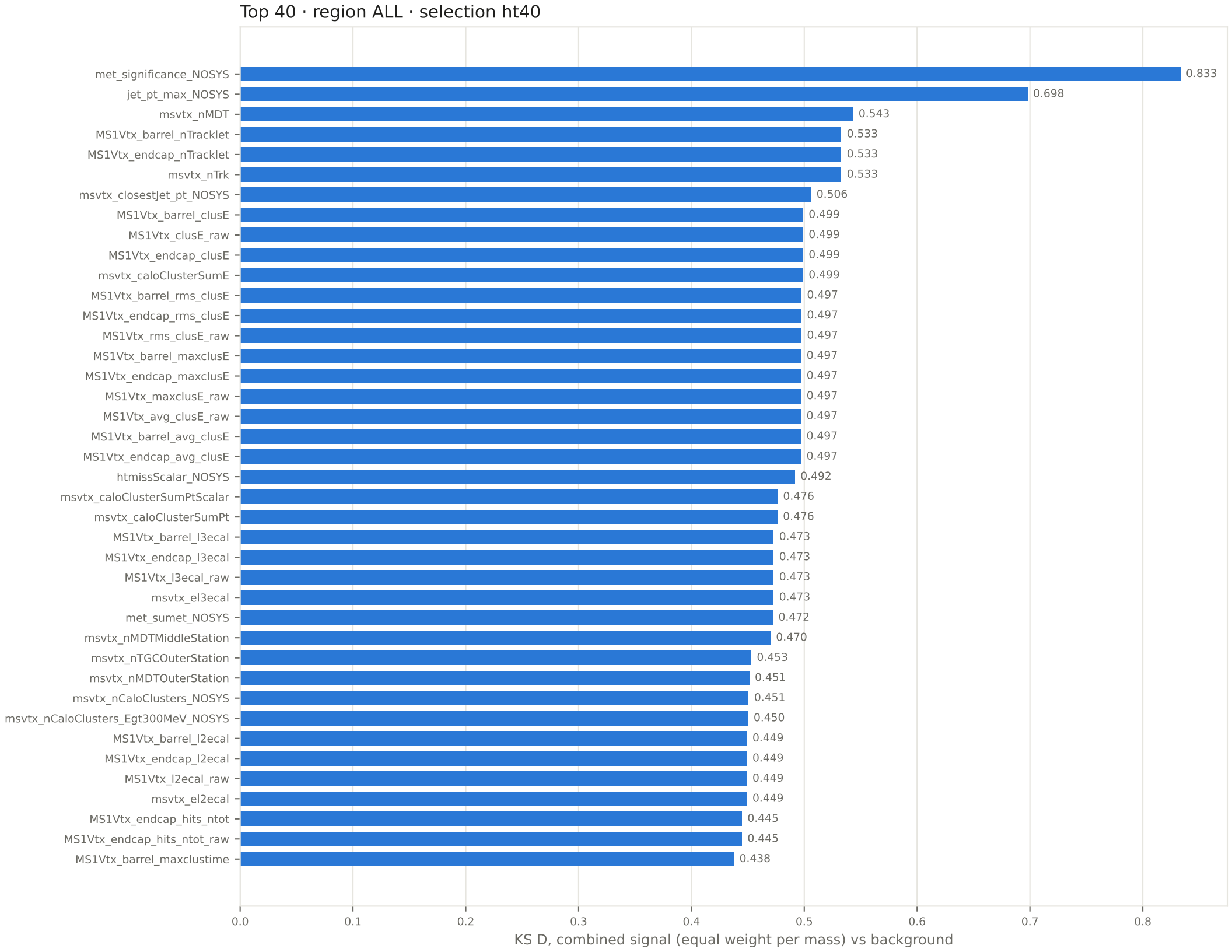
Top 40 table · region ENDCAP · selection ht40

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	met_significance_NOSYS	0.827	0.859	0.746	0.925	7,390
2	msvtx_nTGCOuterStation	0.706	0.771	0.589	0.828	7,390
3	MS1Vtx_endcap_hits_ntot	0.700	0.778	0.524	0.856	7,390
4	MS1Vtx_endcap_hits_ntot_raw	0.700	0.778	0.524	0.856	7,390
5	jet_pt_max_NOSYS	0.687	0.746	0.539	0.861	7,390
6	msvtx_nTGCExtendedStation	0.683	0.753	0.567	0.819	7,390
7	msvtx_nTGC	0.681	0.758	0.562	0.829	7,390
8	msvtx_nMDT	0.654	0.745	0.457	0.838	7,390
9	msvtx_nMDTMiddleStation	0.631	0.725	0.481	0.821	7,390
10	MS1Vtx_barrel_nTracklet	0.594	0.595	0.539	0.631	7,390
11	MS1Vtx_endcap_nTracklet	0.594	0.595	0.539	0.631	7,390
12	msvtx_nTrk	0.594	0.595	0.539	0.631	7,390
13	msvtx_nTGCMiddleStation	0.589	0.673	0.445	0.746	7,390
14	msvtx_nMDTOuterStation	0.569	0.623	0.446	0.678	7,390
15	msvtx_closestJet_pt_NOSYS	0.498	0.519	0.462	0.550	7,390
16	msvtx_chiSquared	0.485	0.565	0.571	1.000	7,390
17	msvtx_caloClusterSumPt	0.477	0.477	0.473	0.492	7,390
18	msvtx_caloClusterSumPtScalar	0.476	0.476	0.473	0.493	7,390
19	htmissScalar_NOSYS	0.469	0.559	0.290	0.781	7,390
20	MS1Vtx_barrel_clusE	0.465	0.469	0.454	0.479	7,390
21	MS1Vtx_clusE_raw	0.465	0.469	0.454	0.479	7,390
22	MS1Vtx_endcap_clusE	0.465	0.469	0.454	0.479	7,390
23	msvtx_caloClusterSumE	0.465	0.469	0.454	0.479	7,390
24	MSeg_endcap_avg_dR	0.462	0.541	0.235	0.666	7,390
25	MSeg_endcap_avg_dR_raw	0.462	0.541	0.235	0.666	7,390
26	MS1Vtx_barrel_maxclusE	0.461	0.464	0.435	0.472	7,390
27	MS1Vtx_endcap_maxclusE	0.461	0.464	0.435	0.472	7,390
28	MS1Vtx_maxclusE_raw	0.461	0.464	0.435	0.472	7,390
29	MS1Vtx_barrel_l3ecal	0.451	0.453	0.435	0.469	7,390
30	MS1Vtx_endcap_l3ecal	0.451	0.453	0.435	0.469	7,390
31	MS1Vtx_l3ecal_raw	0.451	0.453	0.435	0.469	7,390
32	msvtx_el3ecal	0.451	0.453	0.435	0.469	7,390
33	met_sumet_NOSYS	0.450	0.548	0.254	0.777	7,390
34	MS1Vtx_barrel_rms_clusE	0.448	0.455	0.425	0.468	7,390
35	MS1Vtx_endcap_rms_clusE	0.448	0.455	0.425	0.468	7,390
36	MS1Vtx_rms_clusE_raw	0.448	0.455	0.425	0.468	7,390
37	msvtx_nCaloClusters_Egt300MeV_NOSYS	0.446	0.446	0.437	0.456	7,390
38	msvtx_nCaloClusters_NOSYS	0.446	0.447	0.440	0.457	7,390
39	MS1Vtx_avg_clusE_raw	0.443	0.452	0.417	0.465	7,390
40	MS1Vtx_barrel_avg_clusE	0.443	0.452	0.417	0.465	7,390


Top 40 · region BARREL · selection ht40
msegUnAssoc_counts/nEML 0.888
muonAllSegments_counts/nEML 0.876
msvtx_nRPCOuterStation 0.868
met_significance_NOSYS 0.833
msvtx_nRPC 0.806
nMSeg_ratio_EIEM 0.758
nMSeg_ratio_EIEM_raw 0.758
nEOL_NOSYS 0.743
jet_pt_max_NOSYS 0.713
muonAllSegments_counts/nEOL 0.711
MS1Vtx_barrel_hits_ntot 0.693
MS1Vtx_barrel_hits_ntot_raw 0.693
msvtx_nRPCMiddleStation 0.593
msvtx_closestJet_pt_NOSYS 0.559
htmissScalar_NOSYS 0.535
met_sumet_NOSYS 0.527
MS1Vtx_barrel_clusE 0.473
MS1Vtx_clusE_raw 0.473
MS1Vtx_endcap_clusE 0.473
msvtx_caloClusterSumE 0.473
msvtx_caloClusterSumPt 0.472
msvtx_caloClusterSumPtScalar 0.471
MS1Vtx_barrel_rms_clusE 0.468
MS1Vtx_endcap_rms_clusE 0.468
MS1Vtx_rms_clusE_raw 0.468
MS1Vtx_barrel_maxclusE 0.457
MS1Vtx_endcap_maxclusE 0.457
MS1Vtx_maxclusE_raw 0.457
MS1Vtx_avg_clusE_raw 0.453
MS1Vtx_barrel_avg_clusE 0.453
MS1Vtx_endcap_avg_clusE 0.453
MSeg_barrel_avg_dR 0.452
MSeg_barrel_avg_dR_raw 0.452
MS1Vtx_barrel_l3ecal 0.444
MS1Vtx_endcap_l3ecal 0.444
MS1Vtx_l3ecal_raw 0.444
msvtx_el3ecal 0.444
msvtx_nCaloClusters_Egt300MeV_NOSYS 0.439
htmiss_y_NOSYS 0.439
msvtx_nCaloClusters_NOSYS 0.438
0.0
0.2
0.4
0.6
0.8
KS D, combined signal (equal weight per mass) vs background

Top 40 table · region BARREL · selection ht40

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	msegUnAssoc_counts/nEML	0.888	0.873	0.851	0.924	220
2	muonAllSegments_counts/nEML	0.876	0.856	0.827	0.929	220
3	msvtx_nRPCOuterStation	0.868	0.895	0.789	0.928	220
4	met_significance_NOSYS	0.833	0.877	0.741	0.926	220
5	msvtx_nRPC	0.806	0.861	0.700	0.922	220
6	nMSeg_ratio_EIEM	0.758	0.751	0.731	0.806	220
7	nMSeg_ratio_EIEM_raw	0.758	0.751	0.731	0.806	220
8	nEOL_NOSYS	0.743	0.683	0.612	0.863	220
9	jet_pt_max_NOSYS	0.713	0.766	0.580	0.859	220
10	muonAllSegments_counts/nEOL	0.711	0.651	0.582	0.848	220
11	MS1Vtx_barrel_hits_ntot	0.693	0.799	0.454	0.893	220
12	MS1Vtx_barrel_hits_ntot_raw	0.693	0.799	0.454	0.893	220
13	msvtx_nRPCMiddleStation	0.593	0.683	0.358	0.766	220
14	msvtx_closestJet_pt_NOSYS	0.559	0.585	0.497	0.601	220
15	htmissScalar_NOSYS	0.535	0.621	0.347	0.796	220
16	met_sumet_NOSYS	0.527	0.632	0.313	0.804	220
17	MS1Vtx_barrel_clusE	0.473	0.474	0.445	0.496	220
18	MS1Vtx_clusE_raw	0.473	0.474	0.445	0.496	220
19	MS1Vtx_endcap_clusE	0.473	0.474	0.445	0.496	220
20	msvtx_caloClusterSumE	0.473	0.474	0.445	0.496	220
21	msvtx_caloClusterSumPt	0.472	0.471	0.449	0.493	220
22	msvtx_caloClusterSumPtScalar	0.471	0.471	0.450	0.491	220
23	MS1Vtx_barrel_rms_clusE	0.468	0.470	0.431	0.491	220
24	MS1Vtx_endcap_rms_clusE	0.468	0.470	0.431	0.491	220
25	MS1Vtx_rms_clusE_raw	0.468	0.470	0.431	0.491	220
26	MS1Vtx_barrel_maxclusE	0.457	0.458	0.423	0.482	220
27	MS1Vtx_endcap_maxclusE	0.457	0.458	0.423	0.482	220
28	MS1Vtx_maxclusE_raw	0.457	0.458	0.423	0.482	220
29	MS1Vtx_avg_clusE_raw	0.453	0.464	0.401	0.480	220
30	MS1Vtx_barrel_avg_clusE	0.453	0.464	0.401	0.480	220
31	MS1Vtx_endcap_avg_clusE	0.453	0.464	0.401	0.480	220
32	MSeg_barrel_avg_dR	0.452	0.583	0.112	0.735	220
33	MSeg_barrel_avg_dR_raw	0.452	0.583	0.112	0.735	220
34	MS1Vtx_barrel_l3ecal	0.444	0.451	0.414	0.476	220
35	MS1Vtx_endcap_l3ecal	0.444	0.451	0.414	0.476	220
36	MS1Vtx_l3ecal_raw	0.444	0.451	0.414	0.476	220
37	msvtx_el3ecal	0.444	0.451	0.414	0.476	220
38	msvtx_nCaloClusters_Egt300MeV_NOSYS	0.439	0.436	0.414	0.474	220
39	htmiss_y_NOSYS	0.439	0.440	0.370	0.470	220
40	msvtx_nCaloClusters_NOSYS	0.438	0.435	0.413	0.472	220


Top 40 · region ALL · selection ht40
met_significance_NOSYS 0.833
jet_pt_max_NOSYS 0.698
msvtx_nMDT 0.543
MS1Vtx_barrel_nTracklet 0.533
MS1Vtx_endcap_nTracklet 0.533
msvtx_nTrk 0.533
msvtx_closestJet_pt_NOSYS 0.506
MS1Vtx_barrel_clusE 0.499
MS1Vtx_clusE_raw 0.499
MS1Vtx_endcap_clusE 0.499
msvtx_caloClusterSumE 0.499
MS1Vtx_barrel_rms_clusE 0.497
MS1Vtx_endcap_rms_clusE 0.497
MS1Vtx_rms_clusE_raw 0.497
MS1Vtx_barrel_maxclusE 0.497
MS1Vtx_endcap_maxclusE 0.497
MS1Vtx_maxclusE_raw 0.497
MS1Vtx_avg_clusE_raw 0.497
MS1Vtx_barrel_avg_clusE 0.497
MS1Vtx_endcap_avg_clusE 0.497
htmissScalar_NOSYS 0.492
msvtx_caloClusterSumPtScalar 0.476
msvtx_caloClusterSumPt 0.476
MS1Vtx_barrel_l3ecal 0.473
MS1Vtx_endcap_l3ecal 0.473
MS1Vtx_l3ecal_raw 0.473
msvtx_el3ecal 0.473
met_sumet_NOSYS 0.472
msvtx_nMDTMiddleStation 0.470
msvtx_nTGCOuterStation 0.453
msvtx_nMDTOuterStation 0.451
msvtx_nCaloClusters_NOSYS 0.451
msvtx_nCaloClusters_Egt300MeV_NOSYS 0.450
MS1Vtx_barrel_l2ecal 0.449
MS1Vtx_endcap_l2ecal 0.449
MS1Vtx_l2ecal_raw 0.449
msvtx_el2ecal 0.449
MS1Vtx_endcap_hits_ntot 0.445
MS1Vtx_endcap_hits_ntot_raw 0.445
MS1Vtx_barrel_maxclustime 0.438
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
KS D, combined signal (equal weight per mass) vs background

Top 40 table · region ALL · selection ht40

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	met_significance_NOSYS	0.833	0.866	0.749	0.924	7,610
2	jet_pt_max_NOSYS	0.698	0.756	0.538	0.857	7,610
3	msvtx_nMDT	0.543	0.651	0.343	0.771	7,610
4	MS1Vtx_barrel_nTracklet	0.533	0.536	0.490	0.557	7,610
5	MS1Vtx_endcap_nTracklet	0.533	0.536	0.490	0.557	7,610
6	msvtx_nTrk	0.533	0.536	0.490	0.557	7,610
7	msvtx_closestJet_pt_NOSYS	0.506	0.529	0.462	0.554	7,610
8	MS1Vtx_barrel_clusE	0.499	0.500	0.496	0.505	7,610
9	MS1Vtx_clusE_raw	0.499	0.500	0.496	0.505	7,610
10	MS1Vtx_endcap_clusE	0.499	0.500	0.496	0.505	7,610
11	msvtx_caloClusterSumE	0.499	0.500	0.496	0.505	7,610
12	MS1Vtx_barrel_rms_clusE	0.497	0.504	0.482	0.512	7,610
13	MS1Vtx_endcap_rms_clusE	0.497	0.504	0.482	0.512	7,610
14	MS1Vtx_rms_clusE_raw	0.497	0.504	0.482	0.512	7,610
15	MS1Vtx_barrel_maxclusE	0.497	0.500	0.479	0.507	7,610
16	MS1Vtx_endcap_maxclusE	0.497	0.500	0.479	0.507	7,610
17	MS1Vtx_maxclusE_raw	0.497	0.500	0.479	0.507	7,610
18	MS1Vtx_avg_clusE_raw	0.497	0.505	0.478	0.512	7,610
19	MS1Vtx_barrel_avg_clusE	0.497	0.505	0.478	0.512	7,610
20	MS1Vtx_endcap_avg_clusE	0.497	0.505	0.478	0.512	7,610
21	htmissScalar_NOSYS	0.492	0.580	0.315	0.784	7,610
22	msvtx_caloClusterSumPtScalar	0.476	0.477	0.465	0.491	7,610
23	msvtx_caloClusterSumPt	0.476	0.477	0.464	0.490	7,610
24	MS1Vtx_barrel_l3ecal	0.473	0.478	0.456	0.491	7,610
25	MS1Vtx_endcap_l3ecal	0.473	0.478	0.456	0.491	7,610
26	MS1Vtx_l3ecal_raw	0.473	0.478	0.456	0.491	7,610
27	msvtx_el3ecal	0.473	0.478	0.456	0.491	7,610
28	met_sumet_NOSYS	0.472	0.570	0.263	0.780	7,610
29	msvtx_nMDTMiddleStation	0.470	0.563	0.302	0.665	7,610
30	msvtx_nTGCOuterStation	0.453	0.494	0.351	0.521	7,610
31	msvtx_nMDTOuterStation	0.451	0.531	0.308	0.625	7,610
32	msvtx_nCaloClusters_NOSYS	0.451	0.452	0.435	0.469	7,610
33	msvtx_nCaloClusters_Egt300MeV_NOSYS	0.450	0.451	0.440	0.467	7,610
34	MS1Vtx_barrel_l2ecal	0.449	0.456	0.424	0.466	7,610
35	MS1Vtx_endcap_l2ecal	0.449	0.456	0.424	0.466	7,610
36	MS1Vtx_l2ecal_raw	0.449	0.456	0.424	0.466	7,610
37	msvtx_el2ecal	0.449	0.456	0.424	0.466	7,610
38	MS1Vtx_endcap_hits_ntot	0.445	0.498	0.317	0.541	7,610
39	MS1Vtx_endcap_hits_ntot_raw	0.445	0.498	0.317	0.541	7,610
40	MS1Vtx_barrel_maxclustime	0.438	0.441	0.412	0.453	7,610

Largest drops all -> ht40 · region ENDCAP

feature	D_all	D_ht40	delta	D_split
htmissScalar_NOSYS	0.789	0.469	-0.319	1.000
msvtx_jetMinDR_NOSYS	0.493	0.178	-0.315	0.622
met_sumet_NOSYS	0.687	0.450	-0.237	0.583
jet_n_NOSYS	0.511	0.277	-0.234	1.000
msvtx_closestJet_pt_NOSYS	0.687	0.498	-0.189	0.543
jet_pt_max_NOSYS	0.854	0.687	-0.167	0.643
msvtx_trackHardMinDR_NOSYS	0.243	0.083	-0.159	0.293
MS1Vtx_barrel_mindR_jetcut	0.308	0.169	-0.139	0.467
MS1Vtx_endcap_mindR_jetcut	0.308	0.169	-0.139	0.467
MS1Vtx_mindR_jetcut_raw	0.308	0.169	-0.139	0.467
track_scalar_sum_pt_endcap	0.286	0.173	-0.113	0.220
track_scalar_sum_pt_endcap_raw	0.286	0.173	-0.113	0.220
track_scalar_sum_pt_barrel	0.213	0.117	-0.095	0.212
track_scalar_sum_pt_barrel_raw	0.213	0.117	-0.095	0.212
mseg_nSeg	0.109	0.058	-0.051	0.112
met_significance_NOSYS	0.876	0.827	-0.049	0.178
mseg_counts/nEOL	0.052	0.014	-0.038	0.076
muonAllSegments_counts/nEIL	0.119	0.083	-0.036	0.060
msegNCB_counts/nEIL	0.151	0.124	-0.027	0.047
muonSegments_nSeg	0.201	0.178	-0.024	0.080
mseg_counts/nBIL	0.041	0.018	-0.023	0.029
msegUnAssoc_counts/nEIL	0.115	0.092	-0.023	0.038
muonAllSegments_counts/nEOL	0.116	0.094	-0.022	0.050
msvtx_nTGC	0.701	0.681	-0.020	0.062
msvtx_nTGCOuterStation	0.725	0.706	-0.018	0.043
msvtx_nMDT	0.672	0.654	-0.018	0.056
MS1Vtx_endcap_hits_ntot	0.718	0.700	-0.018	0.067
MS1Vtx_endcap_hits_ntot_raw	0.718	0.700	-0.018	0.067
msvtx_nTGCExtendedStation	0.700	0.683	-0.017	0.040
msvtx_nMDTMiddleStation	0.648	0.631	-0.016	0.050
muonAllSegments_counts/nEML	0.179	0.163	-0.016	0.043
msvtx_nMDTOuterStation	0.583	0.569	-0.014	0.032
msegUnAssoc_nSeg	0.187	0.175	-0.012	0.050
msvtx_nTGCMiddleStation	0.601	0.589	-0.012	0.054
MS1Vtx_barrel_nTracklet	0.606	0.594	-0.011	0.015
MS1Vtx_endcap_nTracklet	0.606	0.594	-0.011	0.015
msvtx_nTrk	0.606	0.594	-0.011	0.015
mseg_counts/nBML	0.053	0.043	-0.010	0.013
nEOL_NOSYS	0.107	0.098	-0.009	0.026
msegUnAssoc_counts/nEML	0.187	0.178	-0.008	0.026

delta = D_ht40 - D_all. Large negative: the separation came from events the cut removes.
D_split = background below vs at/above the cut (shape change across the cut).

Largest drops all -> ht40 · region BARREL

feature	D_all	D_ht40	delta	D_split
msvtx_jetMinDR_NOSYS	0.555	0.166	-0.389	0.703
jet_n_NOSYS	0.562	0.236	-0.326	1.000
htmissScalar_NOSYS	0.798	0.535	-0.263	1.000
met_sumet_NOSYS	0.737	0.527	-0.210	0.638
MS1Vtx_barrel_mindR_jetcut	0.365	0.180	-0.185	0.492
MS1Vtx_endcap_mindR_jetcut	0.365	0.180	-0.185	0.492
MS1Vtx_mindR_jetcut_raw	0.365	0.180	-0.185	0.492
msvtx_trackHardMinDR_NOSYS	0.320	0.150	-0.170	0.371
jet_pt_max_NOSYS	0.873	0.713	-0.160	0.650
msvtx_closestJet_pt_NOSYS	0.707	0.559	-0.149	0.546
track_scalar_sum_pt_barrel	0.265	0.162	-0.102	0.236
track_scalar_sum_pt_barrel_raw	0.265	0.162	-0.102	0.236
track_scalar_sum_pt_endcap	0.284	0.190	-0.094	0.239
track_scalar_sum_pt_endcap_raw	0.284	0.190	-0.094	0.239
met_significance_NOSYS	0.898	0.833	-0.065	0.230
msegNCB_counts/nEIL	0.168	0.112	-0.056	0.091
msvtx_phi_NOSYS	0.285	0.236	-0.049	0.088
muonAllSegments_counts/nEIL	0.132	0.088	-0.044	0.107
MS1Vtx_barrel_nTracklet	0.442	0.404	-0.038	0.055
MS1Vtx_endcap_nTracklet	0.442	0.404	-0.038	0.055
msvtx_nTrk	0.442	0.404	-0.038	0.055
mseg_counts/nBML	0.135	0.098	-0.036	0.092
MSVtx_MET_barrel_dphi	0.429	0.398	-0.031	0.088
MSVtx_MET_dphi_raw	0.429	0.398	-0.031	0.088
MSVtx_MET_endcap_dphi	0.429	0.398	-0.031	0.088
msegUnAssoc_counts/nEIL	0.136	0.108	-0.028	0.106
msvtx_nTGCInnerStation	0.038	0.012	-0.026	0.032
mseg_counts/nBOL	0.132	0.108	-0.025	0.063
msvtx_x	0.087	0.067	-0.021	0.084
msvtx_nTGC	0.108	0.089	-0.019	0.060
msvtx_y	0.166	0.156	-0.010	0.118
msvtx_nTGCInwards	0.007	0.002	-0.005	0.008
msvtx_nTGCExtendedStation	0.034	0.030	-0.004	0.027
msvtx_nTGCMiddleStation	0.059	0.056	-0.003	0.042
msvtx_nMDTExtendedStation	0.222	0.219	-0.002	0.034
globalTriggerMatch_NOSYS	0.000	0.000	-0.000	nan
trigPassed_HLT_e140_lhloose_L1eEM26M	0.000	0.000	-0.000	nan
trigPassed_HLT_e26_lhtight_ivarloose_L1eEM26M	0.000	0.000	-0.000	nan
trigPassed_HLT_e300_etcut_L1eEM26M	0.008	0.008	-0.000	nan
trigPassed_HLT_e60_lhmedium_L1eEM26M	0.000	0.000	-0.000	nan

delta = D_ht40 - D_all. Large negative: the separation came from events the cut removes.
D_split = background below vs at/above the cut (shape change across the cut).

Largest drops all -> ht40 · region ALL

feature	D_all	D_ht40	delta	D_split
msvtx_jetMinDR_NOSYS	0.504	0.177	-0.327	0.624
htmissScalar_NOSYS	0.789	0.492	-0.297	1.000
jet_n_NOSYS	0.525	0.264	-0.261	1.000
met_sumet_NOSYS	0.702	0.472	-0.229	0.584
msvtx_trackHardMinDR_NOSYS	0.275	0.061	-0.214	0.294
msvtx_closestJet_pt_NOSYS	0.689	0.506	-0.183	0.543
MS1Vtx_barrel_mindR_jetcut	0.324	0.160	-0.164	0.467
MS1Vtx_endcap_mindR_jetcut	0.324	0.160	-0.164	0.467
MS1Vtx_mindR_jetcut_raw	0.324	0.160	-0.164	0.467
jet_pt_max_NOSYS	0.856	0.698	-0.159	0.643
track_scalar_sum_pt_endcap	0.287	0.178	-0.109	0.220
track_scalar_sum_pt_endcap_raw	0.287	0.178	-0.109	0.220
track_scalar_sum_pt_barrel	0.223	0.127	-0.096	0.212
track_scalar_sum_pt_barrel_raw	0.223	0.127	-0.096	0.212
met_significance_NOSYS	0.882	0.833	-0.049	0.179
muonAllSegments_counts/nEIL	0.095	0.064	-0.031	0.061
msegNCB_counts/nEIL	0.129	0.100	-0.029	0.048
mseg_nSeg	0.084	0.059	-0.025	0.114
mseg_counts/nBIL	0.155	0.132	-0.023	0.029
msegUnAssoc_counts/nEIL	0.105	0.088	-0.017	0.040
MS1Vtx_endcap_hits_ntot	0.462	0.445	-0.017	0.066
MS1Vtx_endcap_hits_ntot_raw	0.462	0.445	-0.017	0.066
msvtx_nMDT	0.560	0.543	-0.017	0.054
msvtx_nTGCOuterStation	0.468	0.453	-0.015	0.042
msvtx_nTGC	0.451	0.437	-0.014	0.060
msvtx_nMDTOuterStation	0.465	0.451	-0.013	0.031
msvtx_nMDTMiddleStation	0.483	0.470	-0.013	0.046
MS1Vtx_barrel_nTracklet	0.545	0.533	-0.012	0.016
MS1Vtx_endcap_nTracklet	0.545	0.533	-0.012	0.016
msvtx_nTrk	0.545	0.533	-0.012	0.016
msvtx_nTGCExtendedStation	0.448	0.436	-0.012	0.039
msvtx_nTGCMiddleStation	0.381	0.371	-0.010	0.053
mseg_counts/nBML	0.242	0.232	-0.010	0.013
mseg_counts/nBOL	0.252	0.245	-0.007	0.009
msvtx_y	0.129	0.123	-0.007	0.011
msvtx_chiSquared	0.405	0.399	-0.006	0.015
msvtx_nTGCInnerStation	0.036	0.031	-0.005	0.012
msvtx_nMDTInnerStation	0.249	0.245	-0.005	0.026
htmiss_y_NOSYS	0.418	0.413	-0.005	0.278
MSVtx_MET_barrel_dphi	0.389	0.385	-0.004	0.046

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